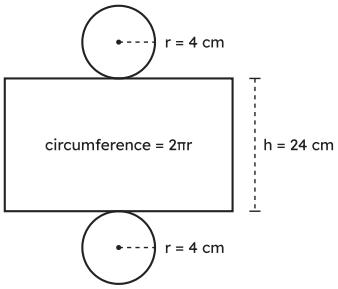
Benchmark: **MA.7.GR.2.1** · Surface area of a cylinder from its net Student code: _____ Calculator: yes

Learning goal: I can find the surface area of a right circular cylinder. (circle your level after scoring)

4	I can explain how the net gives the formula and find surface area with a missing dimension.
3	I can find the surface area of a cylinder from its radius and height.
2	I can identify the parts of a cylinder's net.
1	I need help to start this benchmark.

1. The net of a right circular cylinder is shown. The radius is 4 centimeters and the height is 24 centimeters.

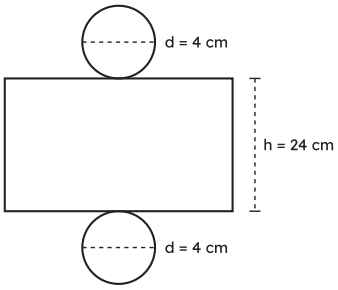
What is the surface area of the cylinder? Write your answer in terms of π.



Not drawn to scale

2. The net of a right circular cylinder is shown. Each circle has a diameter of 4 centimeters, and the rectangle is 24 centimeters tall.

What is the surface area of the cylinder? Write your answer in terms of π.

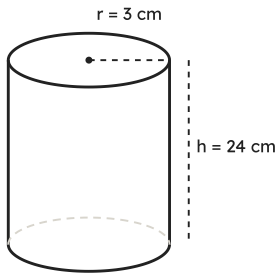


Not drawn to scale

Turn over →

3. The right circular cylinder shown has a radius of 3 centimeters and a height of 24 centimeters.

Which net, with its dimensions, matches the cylinder?



A. Two circles of radius 3 cm and a 3π cm by 24 cm rectangle

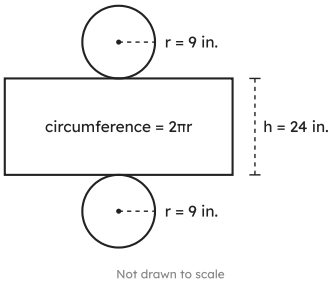
B. Two circles of radius 3 cm and a 6 cm by 24 cm rectangle

C. Two circles of radius 3 cm and a 6π cm by 24 cm rectangle

D. One circle of radius 3 cm and a 6π cm by 24 cm rectangle

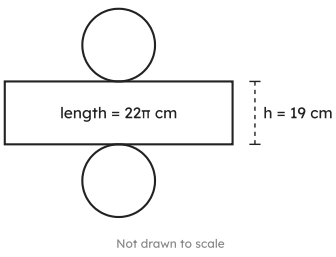
4. The net of a right circular cylinder is shown. The circles have a radius of 9 inches and the rectangle is 24 inches tall.

What is the area of the rectangle, in terms of π?



5. The net of a right circular cylinder is shown. The rectangle is 22π centimeters long and 19 centimeters tall.

What is the surface area of the cylinder? Write your answer in terms of π.



Score: ____ / 5 My level: 1 2 3 4 → copy both to your progress strip

Answer key — C-GR.2.1 · MA.7.GR.2.1 · Surface area of a cylinder from its net

Unit 12: Surface Area and Volume. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **224π square centimeters**

Two bases: $2 \times \pi \times 4^2 = 32\pi$.
The rectangle: $2\pi \times 4 \times 24 = 192\pi$.
Total: $32\pi + 192\pi = 224\pi$ square centimeters.

2. Regular **104π square centimeters**

The radius is $4 \div 2 = 2$ centimeters.
Two circles: $2 \times \pi(2)^2 = 8\pi$.
The rectangle wraps around a circle, so its length is the circumference $2\pi(2) = 4\pi$ centimeters: $4\pi \times 24 = 96\pi$.
Surface area: $8\pi + 96\pi = 104\pi$ square centimeters.

3. Regular **C. Two circles of radius 3 cm and a 6π cm by 24 cm rectangle**

A cylinder has two circular bases, so the net has two circles of radius 3 cm.
The rectangle wraps around a base, so its length is the circumference, $2\pi(3) = 6\pi$ cm, and its width is the height, 24 cm.

4. Regular **432π square inches**

The rectangle wraps around the circle, so its width is the circumference: $2\pi r = 18\pi$ inches.
$\text{Area} = 18\pi \times 24 = 432\pi$ square inches.

5. Rigor **660π square centimeters**

The rectangle wraps around a base, so its length is the circumference of a circle: $2\pi r = 22\pi$, so $r = 22 \div 2 = 11$ centimeters.
Two circles: $2 \times \pi(11)^2 = 242\pi$.
Rectangle: $22\pi \times 19 = 418\pi$.
Surface area: $242\pi + 418\pi = 660\pi$ square centimeters.

Re-check strip (3 items, C-GR.2.1-R)

1. Regular **168π square centimeters**

Two bases: $2 \times \pi \times 6^2 = 72\pi$.
The rectangle: $2\pi \times 6 \times 8 = 96\pi$.
Total: $72\pi + 96\pi = 168\pi$ square centimeters.

2. Regular **160π square centimeters**

The radius is $8 \div 2 = 4$ centimeters.
Two circles: $2 \times \pi(4)^2 = 32\pi$.
The rectangle wraps around a circle, so its length is the circumference $2\pi(4) = 8\pi$ centimeters: $8\pi \times 16 = 128\pi$.
Surface area: $32\pi + 128\pi = 160\pi$ square centimeters.

3. Regular **84π square inches**

The rectangle wraps around the circle, so its width is the circumference: $2\pi r = 4\pi$ inches.
$\text{Area} = 4\pi \times 21 = 84\pi$ square inches.

Comparison strip (2 items, C-GR.2.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **182π square inches**

The rectangle wraps around the circle, so its width is the circumference: $2\pi r = 14\pi$ inches.
$\text{Area} = 14\pi \times 13 = 182\pi$ square inches.

2. Regular **414π square centimeters**

The radius is $18 \div 2 = 9$ centimeters.
Two circles: $2 \times \pi(9)^2 = 162\pi$.
The rectangle wraps around a circle, so its length is the circumference $2\pi(9) = 18\pi$ centimeters: $18\pi \times 14 = 252\pi$.
Surface area: $162\pi + 252\pi = 414\pi$ square centimeters.

Reteach plan — C-GR.2.1

MA.7.GR.2.1 · Surface area of a cylinder from its net

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

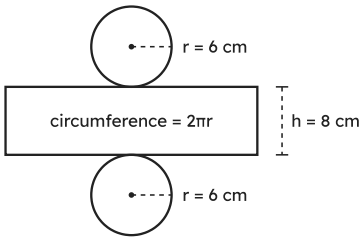
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.2.1-R

Benchmark: **MA.7.GR.2.1** · Surface area of a cylinder from its net Student code: _____ Calculator: yes

1. The net of a right circular cylinder is shown. The radius is 6 centimeters and the height is 8 centimeters.

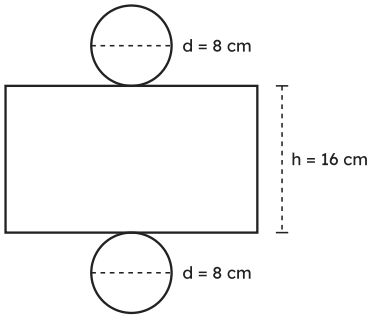
What is the surface area of the cylinder? Write your answer in terms of π.



Not drawn to scale

2. The net of a right circular cylinder is shown. Each circle has a diameter of 8 centimeters, and the rectangle is 16 centimeters tall.

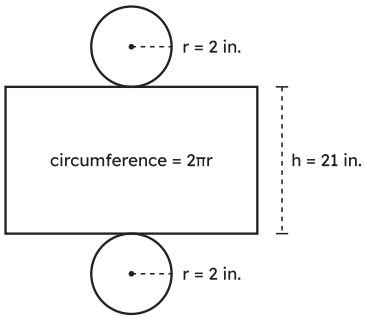
What is the surface area of the cylinder? Write your answer in terms of π.



Not drawn to scale

3. The net of a right circular cylinder is shown. The circles have a radius of 2 inches and the rectangle is 21 inches tall.

What is the area of the rectangle, in terms of π?



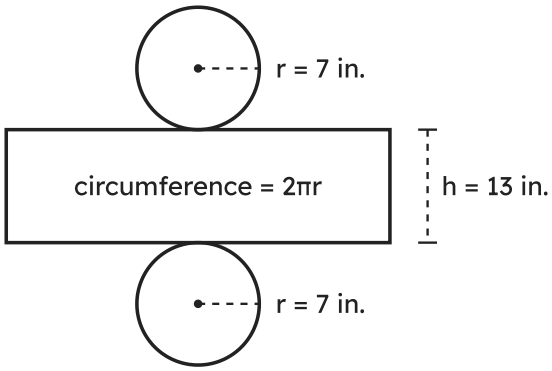
Not drawn to scale

Score: ____ / 3 → **copy to your progress strip**

Benchmark: **MA.7.GR.2.1** · Surface area of a cylinder from its net Student code: _____ Calculator: yes

1. The net of a right circular cylinder is shown. The circles have a radius of 7 inches and the rectangle is 13 inches tall.

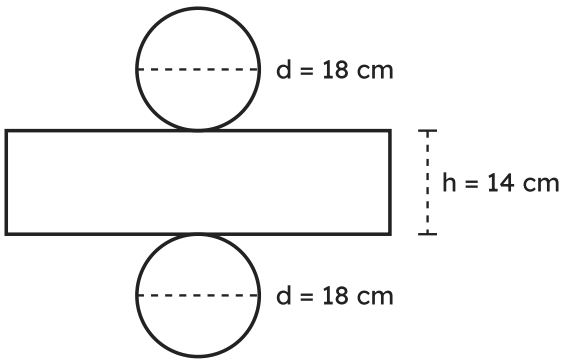
What is the area of the rectangle, in terms of π?



Not drawn to scale

2. The net of a right circular cylinder is shown. Each circle has a diameter of 18 centimeters, and the rectangle is 14 centimeters tall.

What is the surface area of the cylinder? Write your answer in terms of π.



Not drawn to scale

Score: ____ / 2 **→ copy to your progress strip**